

LJUBLJANA BEZIGRAD, Slovenia

WMO: 140150

Lat: 46.0656N

Lon: 14.5122E

Elev: 298

StdP: 97.80

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.4	-6.5	-13.3	1.2	-5.9	-10.8	1.5	-3.1	5.5	5.0	4.8	4.6	1.2	50	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.0	21.2	30.4	20.5	28.8	19.8	22.0	30.1	21.2	28.6	20.4	27.3	2.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.7	25.3	18.6	13.9	24.4	17.9	13.3	23.5	65.8	30.2	62.6	28.8	59.8	27.3	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.2	4.3	3.6	-11.2	34.6	2.8	1.9	-13.2	35.9	-14.8	37.1	-16.4	38.1	-18.4	39.5		
			-11.8	23.5	2.7	1.2	-13.7	24.4	-15.3	25.1	-16.8	25.7	-18.8	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	1.0	2.7	7.1	11.7	16.1	20.0	21.8	21.5	16.5	11.8	6.7	1.6
	DBStd	8.25	3.93	3.85	3.89	3.60	3.22	3.47	3.03	3.15	3.26	3.79	4.27	4.12
	HDD10.0	1017	281	204	103	24	1	0	0	0	0	25	115	263
	HDD18.3	2789	539	437	348	200	83	23	7	8	73	202	351	519
	CDD10.0	1597	0	1	13	75	191	300	367	356	196	82	14	1
	CDD18.3	328	0	0	0	1	15	73	116	105	19	1	0	0
	CDH23.3	2772	0	0	0	14	138	622	1012	870	112	2	0	0
	CDH26.7	894	0	0	0	1	19	192	356	309	17	0	0	0
Wind	WSAvg	1.4	1.2	1.4	1.7	1.7	1.6	1.5	1.5	1.4	1.3	1.3	1.2	1.1
Precipitation	PrecAvg	1362	66	75	87	98	110	140	128	133	146	143	141	95
	PrecMax	1867	191	273	196	154	205	208	242	266	370	402	343	231
	PrecMin	1025	5	2	9	3	40	53	44	24	40	2	1	0
	PrecStd	190	42	62	53	39	41	38	48	62	79	86	78	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	15.6	20.7	25.4	28.8	32.6	34.4	34.6	28.7	23.1	17.4	13.5
		MCWB	8.6	8.2	11.5	15.0	18.9	21.4	21.8	21.7	19.7	17.0	12.9	10.9
	2%	DB	9.9	12.1	17.9	22.6	26.5	30.7	32.1	32.0	26.4	20.9	15.2	11.1
		MCWB	7.5	7.4	10.4	13.5	17.3	20.6	21.2	21.1	18.9	16.0	12.4	9.2
	5%	DB	8.5	10.3	15.6	20.6	24.8	29.0	30.3	30.1	24.3	19.1	13.7	9.3
		MCWB	6.6	6.8	9.4	12.6	16.4	19.7	20.5	20.4	17.7	14.9	11.7	7.7
	10%	DB	7.1	8.5	13.5	18.5	22.9	27.2	28.6	28.1	22.3	17.4	12.5	7.4
		MCWB	5.4	5.6	8.5	11.8	15.5	18.9	19.8	19.7	16.7	14.2	10.7	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.6	12.3	15.7	19.7	22.6	23.2	23.0	20.4	17.6	14.0	11.2
		MCDB	10.9	12.7	18.6	23.5	27.4	30.7	31.7	32.1	26.8	21.5	15.9	12.9
	2%	WB	8.1	8.3	11.1	14.2	18.1	21.4	22.0	21.9	19.3	16.6	12.9	9.5
		MCDB	9.5	11.4	16.0	20.9	24.9	29.0	30.1	29.9	25.2	20.0	14.5	11.1
	5%	WB	6.9	7.2	10.2	13.2	17.0	20.4	21.2	21.0	18.4	15.6	11.9	7.9
		MCDB	8.4	10.1	14.3	19.1	23.4	27.5	28.7	28.2	23.2	18.4	13.7	9.3
	10%	WB	5.6	6.0	9.3	12.2	16.1	19.3	20.3	20.2	17.4	14.6	10.9	6.1
		MCDB	7.0	8.4	12.9	17.3	21.8	25.8	27.2	26.8	21.2	17.1	12.4	7.3
Mean Daily Temperature Range	5% DB	MDBR	5.1	7.3	9.2	9.7	10.2	10.4	10.9	10.7	9.2	7.7	5.2	4.6
		MCDBR	7.1	11.3	13.8	13.9	13.6	13.2	13.6	13.9	12.0	9.8	7.0	6.7
		MCWBR	4.9	6.8	7.2	6.3	5.8	5.2	5.0	5.0	5.3	5.2	4.5	5.0
	5% WB	MCDBR	5.7	10.0	11.4	12.1	12.4	12.4	12.5	12.5	10.9	8.2	5.7	5.9
		MCWBR	4.2	6.3	6.3	5.7	5.5	5.3	5.0	4.9	5.1	4.7	4.1	4.6
Clear-Sky Solar Irradiance	taub	0.325	0.353	0.400	0.450	0.449	0.457	0.453	0.447	0.425	0.400	0.359	0.324	
	taud	2.397	2.315	2.212	2.099	2.147	2.175	2.196	2.225	2.250	2.294	2.360	2.441	
	Ebn at Noon	767	811	818	807	822	816	813	802	783	743	711	728	
	Edh at Noon	73	96	122	150	147	144	140	131	117	97	76	64	
All-Sky Solar Radiation	RadAvg	1.25	2.05	3.22	4.25	5.24	5.93	5.98	5.23	3.67	2.37	1.31	1.05	
	RadStd	0.19	0.34	0.45	0.51	0.52	0.40	0.39	0.58	0.48	0.28	0.20	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (69 neighbors)	+0.49	N/A	N/A	+1.17	+0.88	+0.70	N/A	-102	+119	+90
		+0.49	N/A	N/A	+0.81	+0.46	+0.36	-80	-127	+95	+62

Nomenclature: See separate page